

Energy-Efficient Deep Learning Without Backpropagation: A Rigorous Hardware-Validated Benchmarking Study of Forward-Only Algorithms

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Abstract. Backpropagation (BP) remains the dominant algorithm for training deep neural networks, yet its high memory and energy costs motivate the search for practical alternatives. We present a rigorous, hardware-validated benchmarking study of three forward-only, BP-free training algorithms, namely Forward-Forward (FF), Cascaded Forward (CaFo), and Mono-Forward (MF), evaluated against a fairly tuned BP baseline on MLP and 3-block CNN architectures across four standard vision benchmarks (MNIST, Fashion-MNIST, CIFAR-10, CIFAR-100). Using a principled comparison framework with identical architectures, universal hyperparameter optimisation via Optuna, and direct hardware-level efficiency measurement via the NVIDIA Management Library (NVML), we find that MF consistently matches or surpasses BP in classification accuracy on its native MLP architectures, achieving up to 41% lower energy consumption and up to 34% faster training on these configurations. We further conduct an empirical re-evaluation of memory efficiency, demonstrating that practical overhead can offset theoretical savings from eliminating the backward pass. These findings are specific to MLP architectures on standard benchmarks; performance on CNNs, transformers, or larger-scale settings remains an open question. Our primary contribution is the benchmarking framework itself and, to our knowledge, the first hardware-validated comparison of MF against a fairly tuned BP baseline under controlled conditions.

Keywords: backpropagation-free learning · forward-only algorithms · energy efficiency · hardware benchmarking · deep neural networks

1 Introduction

Deep neural networks (DNNs) have achieved transformative success across AI [12, 5, 3], yet this progress rests on an escalating and unsustainable energy budget. As models scale, their operational requirements are projected to consume between 3% and 4% of global electricity by 2030 [28].

Training costs can be substantial: [27] estimated that training a single large Transformer with neural architecture search produces roughly 284 t of CO₂e, comparable to the lifetime emissions of five average American cars, while [19] reported that training GPT-3 produced approximately 552 t of CO₂e.

Among the algorithmic contributors to training energy costs, backpropagation (BP) [23] plays a central role. BP’s dependence on a full backward pass necessitates storing all intermediate activations, leading to memory costs proportional to network depth. Its sequential, layer-locked nature imposes a “backward locking” bottleneck that constrains parallelisation and prolongs training [8].

These limitations motivate a search for alternatives drawing partial inspiration from biological neural systems: local learning rules, the absence of a global backward error signal, and layer-by-layer credit assignment. None of the algorithms studied here claim full biological plausibility [15]; the **FF** algorithm [6] was explicitly motivated by these considerations, and **MF**’s local projection mechanism shares structural similarities with local learning rules. We note these connections as motivating context rather than as validated claims.

This paper investigates three forward-only algorithms that partially instantiate such local principles:

- **Forward-Forward (FF)** [6], a foundational proof of concept demonstrating the viability of local, forward-only learning via a contrastive “goodness” objective.
- **Cascaded Forward (CaFo)** [30], a structured evolution employing block-wise local classifiers to provide more direct supervisory signals.
- **Mono-Forward (MF)** [4], a refinement using per-layer projection matrices and local cross-entropy losses to improve accuracy and training efficiency.

While the potential of these algorithms is recognised in the literature, a comprehensive and fair evaluation based on direct hardware-level energy metrics has been insufficiently explored. Existing comparisons often overlook architectural differences or inconsistent tuning practices, which can confound algorithmic comparisons [2]. This paper addresses this gap by conducting a systematic, hardware-validated benchmarking comparison of **FF**, **CaFo**, and **MF** against fairly tuned **BP** baselines on their respective native architectures. This benchmarking and hardware characterisation contribution is directly relevant to the PPAM focus on parallel and applied computing.

Contributions. The main contributions of this work are:

- **A Rigorous Comparative Benchmarking Framework:** We establish a transparent, reproducible framework mandating identical native architectures, universal hyperparameter tuning applied equally to all methods, and direct hardware-level efficiency measurement via NVML. This framework is itself a transferable contribution for future BP-free evaluations.
- **Hardware-Validated Benchmarking of MF:** To our knowledge, the first hardware-validated comparison of the Mono-Forward algorithm against a fairly tuned **BP** baseline under controlled conditions, finding that **MF** matches

or surpasses BP in accuracy on MLP architectures while achieving up to 41% lower energy consumption on certain configurations.

- **Empirical Re-evaluation of Memory Efficiency:** We demonstrate that practical implementation overhead can offset the theoretical memory savings from eliminating the backward pass, correcting a common assumption in the BP-free literature.

2 BP-Free Training Algorithms

The high memory cost, sequential backward locking, and lack of biological plausibility of backpropagation have motivated a search for alternative training paradigms [15, 8]. This section describes the three algorithms evaluated in this work.

Forward-Forward (FF). Proposed by Geoffrey Hinton, the FF algorithm [6] replaces the forward and backward passes of BP with two forward passes: one using “positive” (real) data and another using “negative” (contrastive) data. Each layer learns locally to maximise a layer-specific “goodness” metric on positive data while minimising it on negative data, enabling purely local credit assignment. By eliminating the global backward pass, FF demonstrated that effective representation learning is possible with local objectives. Its contrastive design partially instantiates the biological principle of Hebbian-like, local synaptic modification [10]. However, generating meaningful negative samples and the indirect goodness signal were identified as practical limitations, often leading to slow convergence and inefficient hardware utilisation [6, 30].

Cascaded Forward (CaFo). The CaFo algorithm [30] introduces a block-wise framework of cascaded neural blocks, each followed by a local auxiliary classifier. This provides direct cross-entropy supervision at multiple network depths, eliminating the need for negative samples. Neural blocks can be randomly initialised and fixed (CaFo-Rand) or pretrained with Direct Feedback Alignment (CaFo-DFA). While CaFo improves upon FF’s indirect supervision, CaFo-Rand suffers from poor feature quality on complex tasks, while CaFo-DFA improves accuracy at a significant computational cost for block pretraining.

Mono-Forward (MF). MF [4] simplifies and enhances the local learning mechanism by equipping each hidden layer with a dedicated, learnable projection matrix. This matrix maps the layer’s activations directly to class-specific goodness scores, upon which a standard cross-entropy loss is computed locally. Both the layer’s primary weights and its projection matrix are updated using only this local loss signal, enabling a single, clean forward pass during training. This mechanism is structurally similar to the local losses framework of [22], which demonstrated that forward-gradient methods augmented with local auxiliary losses scale favourably. MF reportedly achieves accuracy matching or surpassing BP on MLP architectures, which motivated the rigorous validation in this work.

3 Related Work

Forward-Only and Local-Loss Methods. The foundational **FF** algorithm [6] has inspired multiple improvements. [18] introduced convolutional channel-wise competitive learning for **FF**, extending it to CNN architectures with improved feature quality, a limitation our experiments confirm for the base **FF** method. [16] proposed collaborative layer objectives that allow layers to share information during **FF** training, reducing the isolation inherent in purely local objectives. Both works report improved accuracy over baseline **FF** on standard benchmarks but do not provide hardware-level efficiency comparisons of the kind conducted here.

[22] demonstrated that forward gradient methods augmented with local auxiliary losses scale well, providing theoretical and empirical grounding for **MF**'s local projection approach; their work focuses on scalability to larger models, whereas we focus on hardware-level efficiency under fair experimental conditions. [10] proposed Hebbian learning as a biologically motivated alternative to **BP**, while [17] demonstrated forward-only fine-tuning of large language models via memory-efficient zeroth-order optimisation (**MeZO**) — a different regime (large pretrained models, gradient estimation) than ours (training from scratch, local deterministic losses), but one that contextualises the broader hardware implications of **BP**-free training.

Predictive Coding. Predictive coding offers another biologically motivated alternative to **BP**. [25] demonstrated that predictive coding networks can serve as associative memory solvers, and [24] contextualise where local learning methods fit within the broader landscape of brain-inspired learning. [20] extended predictive coding beyond Gaussian assumptions, while [26] introduced a stable, fully automatic variant achieving competitive performance across diverse tasks. Unlike the methods we evaluate, predictive coding networks maintain explicit generative models of their inputs, incurring different computational trade-offs.

Local Loss Optimisation Theory. [7] analysed local loss optimisation in the infinite-width limit, providing theoretical grounding for why greedy layer-wise training can generalise well. Their analysis supports the hypothesis (Section 6.1) that **MF**'s layer-wise training may act as a form of regularisation, though we treat this as a hypothesis requiring further empirical investigation rather than a confirmed result.

Hardware and Energy Considerations. **BP**-optimised hardware accelerators, including GPUs and TPUs [9], are designed around the forward-backward pass structure of **BP**. [27] and [19] provide empirical grounding for the energy costs of current **BP**-based training, situating our efficiency comparisons within a broader sustainability context.

Positioning This Work. The present study complements the above by providing what is, to our knowledge, the first hardware-validated benchmarking comparison

of FF, CaFo, and MF under a strictly controlled fairness framework: identical architectures, universal hyperparameter tuning, and direct NVML energy measurement. We do not introduce new algorithms; our contribution is evaluative.

4 Framework for Fair and Rigorous Comparison

As a key contribution, we propose a rigorous benchmarking framework designed to isolate the impact of the training algorithm from confounding variables. Throughout, a *hardware-validated comparison* is one in which every efficiency claim is backed by a physical quantity read from the GPU itself while the workload runs — energy, elapsed time and memory occupancy — rather than by an analytical proxy such as a FLOP or parameter count.

Fairness Protocols. A strict set of protocols was adhered to for all experiments:

1. **Native Architecture Replication:** Each algorithm was implemented on its native architecture as described in its source publication. FF and MF were evaluated on their designated MLP structures; CaFo on its defined 3-block CNN. CIFAR inputs are flattened to 1-D vectors for MLP experiments, following the original MF protocol.
2. **Identical Architectures for BP Baselines:** For each experiment, a BP baseline was constructed using an *identical* network architecture, omitting only algorithm-specific components (e.g., goodness heads, projection matrices).
3. **Systematic and Universal Hyperparameter Tuning:** Optuna [1] was applied to *all* algorithms, namely FF, CaFo, MF, and their BP baselines, for every configuration, maximising validation set accuracy. Full Optuna search spaces are in the supplementary material.
4. **Consistent Early Stopping:** Training was governed by a consistent early stopping protocol based on validation performance [21].
5. **Fixed Random Seeds:** All sources of randomness were fixed. Three independent runs used seeds 42, 123, and 7. Full seeding and augmentation details are in the supplementary material.

Note on Baseline Completeness. The absence of a BP baseline augmented with per-layer auxiliary classification heads (BP-DS) is a key limitation of the current design; the full discussion and implications are given in Limitation 6 (Section 6.4).

Architecture and Optimiser Configuration. All layer weights and projection matrices were initialised with Kaiming uniform initialisation. BP baselines were trained with AdamW, while CaFo predictors and MF layers used Adam on a local cross-entropy loss. Each CaFo block comprises a 3×3 convolution, ReLU, 2×2 max-pooling and batch normalisation, with 32, 128 and 512 output channels respectively; the three block-wise predictors are aggregated by summation to form the final prediction. FF used a fixed goodness threshold $\theta = 2.0$ with peer normalisation (factor 0.03, momentum 0.9). MF used a weight decay fixed at 0.0, leaving the learning rate and the number of epochs per layer as the tuned

quantities. Optuna searched the remaining learning rates and weight decays for every algorithm and every BP baseline; the search spaces and selected values are in the supplementary material.

Experimental Setup. Experiments were conducted on MNIST [13], Fashion-MNIST [29], CIFAR-10, and CIFAR-100 [11]. All experiments ran on NVIDIA A100-SXM4-40GB GPUs (Athena cluster, ACK Cyfronet AGH), hosted on AMD EPYC 7742 nodes running Rocky Linux 9.8 with NVIDIA driver 595.71.05, Python 3.10.4 and PyTorch v2.4.0. The system CUDA toolkit version was 12.4.0; the PyTorch wheel (`torch==2.4.0+cu121`) bundles a CUDA 12.1 runtime. The driver version is stated because it governs NVML power-reading semantics; NVML energy readings are obtained via the system driver and are not affected by the runtime version difference. Full environment details are in the supplementary material.

For CIFAR-10 and CIFAR-100, standard training augmentation was applied consistently to all algorithms *and* their BP baselines: random horizontal flips ($p=0.5$) and random 32×32 crops with 4-pixel padding. CIFAR inputs for MLP architectures were flattened to 1-D vectors following the MF protocol [4], with augmentation applied before flattening. No augmentation was applied to MNIST or Fashion-MNIST (see the supplementary material for full details).

Evaluation Metrics. Energy and memory were captured with NVML, the vendor-supplied library that exposes the GPU’s own power, clock and memory telemetry to a host process; it was chosen because it reports what the device actually drew and occupied, so no modelling assumption about the workload enters the efficiency figures. The full metric suite is:

- **Performance:** Test-set classification accuracy (mean $\pm$ std across three independent runs).
- **Efficiency (Direct Measurement):** Training time (s), energy consumption (Wh, via NVML), and peak GPU memory (MiB, via NVML). Exact NVML API usage is in the supplementary material.
- **Estimated CO₂e and GFLOPs (Repository Only):** Available at the public repository; omitted from the main paper due to methodological dependencies that would require separate treatment.

5 Experiments and Results

The experiments reveal a progression among BP-free algorithms with distinct accuracy–efficiency trade-offs, culminating in MF’s favourable profile on MLP architectures. Results are reported as mean $\pm$ standard deviation across three independent runs (seeds 42, 123, 7).

Table 1. Performance and efficiency: FF vs. BP on the Fashion-MNIST 4×2000 MLP (mean $\pm$ std, 3 runs; seeds 42, 123, 7). Better result in each column is shown in **bold**. Time and Energy cover full training to early stopping. Memory is peak GPU allocation reported by NVML.

Algorithm	Accuracy (%)	Time (s)	Energy (W h)	Peak Mem (MiB)
FF-AdamW	89.63 ± 0.14	574.60 ± 22.4	14.28 ± 0.57	1190 ± 8
BP Baseline	88.88 ± 0.08	43.09 ± 1.2	1.48 ± 0.05	1168 ± 4

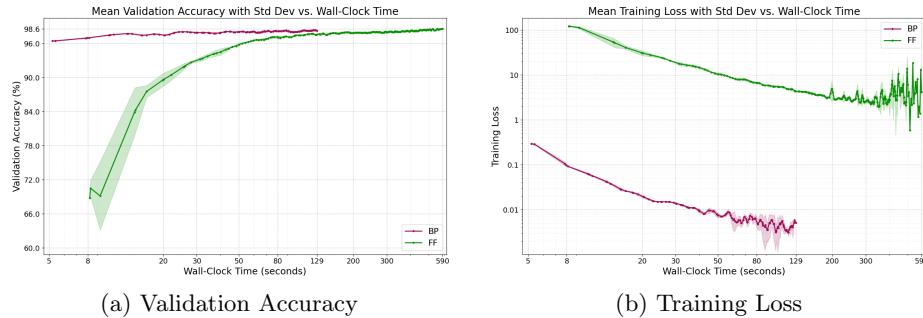


Fig. 1. Convergence dynamics of FF vs. BP on the MNIST 4×2000 MLP. FF’s validation accuracy rises substantially more slowly than BP’s, and exhibits an erratic, elevated training loss compared to BP’s smooth, monotonic decrease, consistent with the goodness metric’s less direct supervision signal. Dataset: MNIST; Architecture: 4×2000 MLP.

5.1 FF: A Conceptually Sound but Computationally Costly Starting Point

The experiments confirmed that FF achieves competitive accuracy on its native MLP architectures, validating its core premise. However, this comes at a prohibitive efficiency cost (Table 1), with slow and volatile convergence causing up to a $13\times$ longer training time (Fashion-MNIST 4×2000 ; see supplementary material) and nearly $10\times$ greater energy use on the same configuration compared to the BP baseline. The accuracy advantage over BP was modest (≤ 0.75 pp) and the efficiency penalty was severe, raising the question of whether the accuracy margin justifies the cost.

The same slow, volatile convergence is visible on MNIST (Figure 1), where FF’s validation accuracy rises substantially more slowly than BP’s, accompanied by an erratic training loss consistent with the indirect goodness supervision signal. The hardware-level cause of the efficiency penalty is revealed by the autopsy of the Fashion-MNIST configuration in Figure 2: volatile GPU clock speeds during FF training indicate inefficient hardware saturation relative to BP’s stable, high utilisation. Counterintuitively, FF consumed slightly *more* peak memory than BP in these tests, refuting a key presumed advantage.

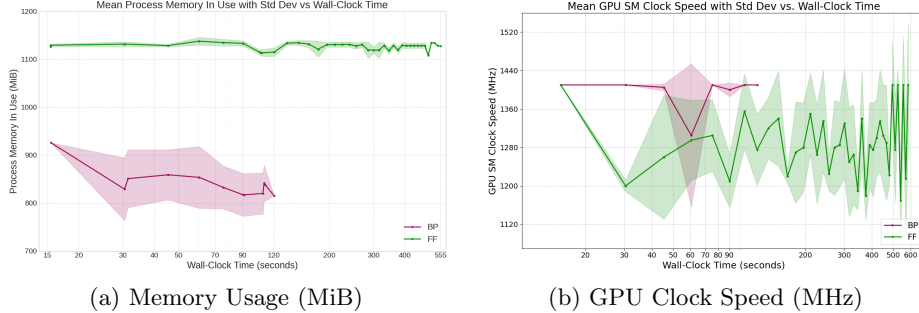


Fig. 2. Hardware-level autopsy of FF vs. BP on the Fashion-MNIST 4×2000 MLP. FF consumes slightly more peak GPU memory than BP, contradicting the theoretical expectation that eliminating the backward pass reduces memory. FF also exhibits volatile GPU clock speeds indicating underutilisation of the compute pipeline, whereas BP maintains stable, high clock rates. Dataset: Fashion-MNIST; Architecture: 4×2000 MLP.

5.2 CaFo: A Structured Alternative with Acute Trade-offs

The CaFo algorithm, evaluated on its native 3-block CNN, revealed a stark accuracy–efficiency trade-off contingent on its feature-learning strategy. The CaFo-Rand-CE variant, using randomly initialised fixed feature extractors, offered modest efficiency gains but at the cost of a significant 13.23 percentage-point accuracy drop on CIFAR-10 relative to BP. The CaFo-DFA-CE variant substantially improved accuracy but proved exceptionally resource-intensive, consuming over four times more energy than BP on CIFAR-10. Table 2 quantifies both sides of this trade-off on that configuration.

Figure 3 illustrates these distinct convergence patterns on Fashion-MNIST. The flat initial segment in the CaFo-DFA validation accuracy curve corresponds to the DFA block-pretraining stage, during which the local predictors are not yet active for the downstream task. This pretraining cost is included in all reported CaFo-DFA time and energy figures, consistently across all datasets.

Table 2. Performance and efficiency: CaFo variants vs. BP on the CIFAR-10 3-block CNN (mean $\pm$ std, 3 runs). Better result in each column is shown in **bold**. Time and Energy cover full training to early stopping and include the CaFo-DFA block-pretraining stage. Memory is peak GPU allocation reported by NVML; it is a device-wide reading that includes the CUDA context and so separates the algorithms only weakly (see Section 6.2). These figures are provisional: they were measured on an earlier revision of the cluster software stack and are pending regeneration in the environment reported above.

Algorithm	Accuracy (%)	Time (s)	Energy (W h)	Peak Mem (MiB)
CaFo-Rand	66.09 $\pm$ 0.32	332.79 $\pm$ 61.94	5.50 $\pm$ 1.00	1014 $\pm$ 0
CaFo-DFA	77.60 $\pm$ 0.25	1314.38 $\pm$ 37.66	27.37 $\pm$ 0.44	1128 $\pm$ 0
BP Baseline	79.32 $\pm$ 3.99	339.51 $\pm$ 18.66	6.81 $\pm$ 0.46	1114 $\pm$ 0

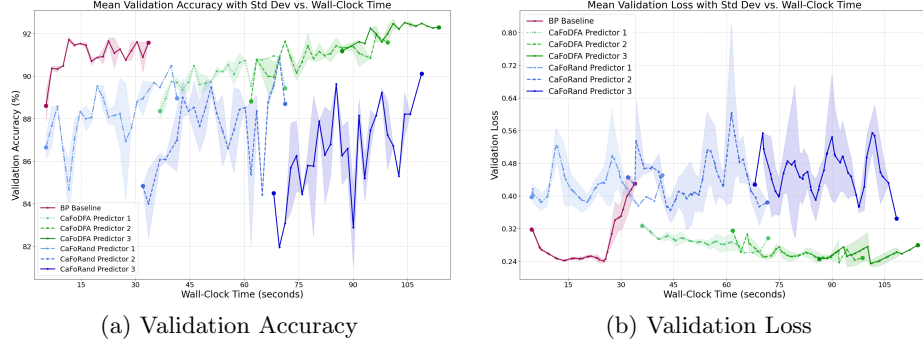


Fig. 3. Convergence dynamics of **CaFo** variants vs. **BP** on the Fashion-MNIST 3-block CNN. **BP** shows smooth, monotonic convergence. **CaFo-DFA** achieves competitive accuracy but incurs an upfront pretraining cost visible as a flat initial segment (the DFA block-pretraining stage), which is included in all reported energy and time figures. **CaFo-Rand** is highly volatile, reflecting the limitations of random fixed features. Dataset: Fashion-MNIST; Architecture: 3-block CNN.

This highlights a critical design insight: while high-quality feature representations are essential for accuracy, the DFA mechanism used by **CaFo-DFA** to learn them is computationally prohibitive. The supplementary material provides full comparative results across all datasets.

5.3 MF: A Superior Accuracy-Efficiency Profile on MLP Architectures

The Mono-Forward algorithm consistently matched or surpassed **BP** classification accuracy on all tested MLP architectures (Table 3). The accuracy advantage was +0.51 pp on Fashion-MNIST and +1.21 pp on CIFAR-10, the most complex tested configuration. Note that CIFAR-10 accuracy for MLP architectures is substantially lower than CNN-based state-of-the-art owing to the elimination of spatial structure via flattening; the comparison here is strictly between an MLP-based **MF** and an MLP-based **BP** baseline on identical architectures. The accuracy margins are modest, and as noted in the statistical note above, with only three runs their statistical robustness cannot be formally established; they are empirical observations, not confirmed claims.

On Fashion-MNIST, **MF** achieved higher accuracy (+0.51 pp) at the cost of modestly increased training time (+18.20%) and energy consumption (+9.90% relative to **BP**), as reported in the supplementary material. This configuration represents an empirical trade-off: higher accuracy is accompanied by slightly higher resource use. On CIFAR-10, the most demanding tested configuration, **MF** was 33.81% faster and consumed 40.78% less energy than **BP** while achieving higher accuracy, representing the strongest case for **MF**'s efficiency advantage.

Hardware monitoring further validates **MF**'s lean design on the CIFAR-10 task. As shown in Figure 4, **MF**'s local training regime results in lower GPU

Table 3. Performance and efficiency: MF vs. BP on the CIFAR-10 3×2000 MLP (mean $\pm$ std, 3 runs; seeds 42, 123, 7). Better result in each column is shown in **bold**. Time and Energy cover full training to early stopping. Memory is peak GPU allocation reported by NVML. CIFAR-10 inputs are flattened to 3072-dimensional vectors following the MF protocol. Peak Mem bold reflects a $\sim 5\%$ margin; see Section 6.2 for discussion.

Algorithm	Accuracy (%)	Time (s)	Energy (W h)	Peak Mem (MiB)
MF	62.34 ± 0.28	177.70 ± 7.1	3.17 ± 0.12	1120 ± 9
BP Baseline	61.13 ± 0.35	268.45 ± 8.1	5.35 ± 0.19	1184 ± 5

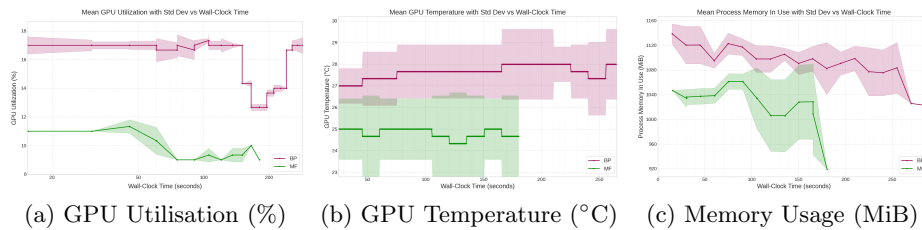


Fig. 4. Hardware monitoring of MF vs. BP on the CIFAR-10 3×2000 MLP. MF’s lower average GPU utilisation (a) and reduced thermal profile (b) are consistent with lower energy consumption. MF’s memory trace (c) remains below BP’s throughout and ends earlier, reflecting its shorter training run; neither trace grows over the run. Dataset: CIFAR-10; Architecture: 3×2000 MLP; GPU: NVIDIA A100-SXM4-40GB.

utilisation and a reduced thermal profile compared to BP. The memory trace in Figure 4(c) shows MF sitting below BP for the whole of training and finishing earlier, consistent with its shorter wall-clock time. Neither trace grows over the course of the run, and the separation between them is small relative to the fixed CUDA context that the reading includes.

Figure 5 illustrates that MF converges to a lower final validation loss than BP’s global end-to-end optimisation on identical architectures, consistent with MF’s accuracy advantage. We discuss a possible regularisation hypothesis to explain this in Section 6.1.

A complete summary comparing the relative performance and efficiency of each algorithm against its fair BP baseline across all tested configurations is presented in the supplementary material.

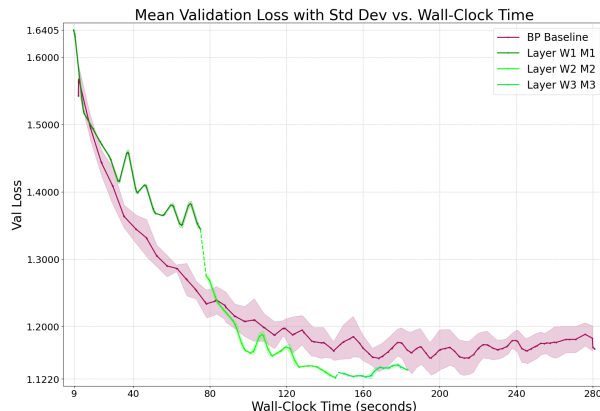


Fig. 5. Validation loss convergence of MF vs. BP on the CIFAR-10 3×2000 MLP. MF’s greedy local optimisations converge to a lower final validation loss, suggesting a possible regularisation effect of layer-wise training (see Section 6.1 for discussion and required diagnostic experiments). Dataset: CIFAR-10; Architecture: 3×2000 MLP.

6 Discussion

We interpret the empirical results, discuss methodological limitations, and draw implications for future work. Where interpretations go beyond the data, we label them explicitly as hypotheses.

6.1 Why MF Succeeds: A Regularisation Hypothesis

FF’s inefficiency traces to its indirect goodness signal and contrastive data generation, resulting in slow convergence and poor GPU utilisation (Section 5). CaFo’s DFA block-pretraining resolves the feature quality problem but at unacceptable energy cost, suggesting that a direct supervisory signal is necessary but that DFA is not the right mechanism. MF’s local projection matrices provide exactly such a direct signal — a local cross-entropy loss at each layer — with minimal overhead. The mechanism is closely related to the local auxiliary losses studied by [22] and [7]. Of particular relevance, [7] analyse local loss optimisation in the infinite-width limit and show that greedy layer-wise training can reduce the effective complexity of the learned representation in ways that support generalisation.

This theoretical result motivates the *regularisation hypothesis* for MF’s accuracy advantage: that the layer-wise, greedy nature of MF’s optimisation implicitly constrains the representational complexity of each layer, guiding the model toward a solution with better generalisation properties than BP’s global, joint optimisation. The lower final validation loss observed for MF in Figure 5 is consistent with this hypothesis. We emphasise that this remains a hypothesis: confirming it would require diagnostic experiments such as measuring the train-validation accuracy gap across training, computing sharpness metrics (e.g., $\lambda_{\max}$ of the loss Hessian), or comparing generalisation bounds under the two optimisation regimes. We leave such analysis to future work.

6.2 Rethinking Memory Efficiency in BP-Free Learning

A critical finding is that eliminating the backward pass does not automatically guarantee superior memory efficiency. While **CaFo-Rand** showed modest memory savings (approximately 6–9% across datasets), both **FF** and **CaFo-DFA** consumed *more* memory than their BP baselines. Even **MF**’s memory advantage on larger MLPs was modest at approximately 5%.

This occurs because the practical overhead of algorithm-specific components, specifically **FF**’s dual data streams, **MF**’s per-layer projection matrices, and **CaFo-DFA**’s DFA feedback tensors, can counteract the theoretical savings from not storing activations for the backward pass. Peak memory as reported by NVML captures GPU-resident tensor allocations; note that PyTorch’s memory allocator may retain freed blocks in a pool, so peak allocation can differ from the maximum simultaneously-live tensor memory. These findings underscore the necessity of direct hardware measurement rather than theoretical analysis alone.

6.3 Implications for Efficient Training and Hardware Co-Design

MF’s energy reduction on CIFAR-10 (approximately 41%) and training time reduction (approximately 34%) relative to a fairly tuned BP baseline are meaningful results for the specific setting studied: MLP architectures on standard vision benchmarks on an NVIDIA A100-SXM4-40GB GPU. Whether these advantages persist on larger networks, different hardware platforms (consumer GPUs, TPUs, edge devices), or other data domains is unknown and represents an important open question.

If forward-only algorithms like **MF** prove effective at larger scales, they could motivate hardware designs with simpler memory subsystems without activation checkpointing and compute pipelines optimised for sequential, layer-local operations, a speculative but well-motivated direction. Extrapolating the present savings to a broader sustainability claim would require demonstrating similar advantages at the scale where energy costs are societally significant [27, 19].

6.4 Limitations

We provide an explicit, comprehensive account of this study’s limitations:

1. **Architecture and scale.** Results apply to 2–4-layer MLP architectures (**FF** and **MF**) and a 3-block CNN (**CaFo**); performance on deeper CNNs, residual networks, transformers, or attention-based architectures is entirely untested. Networks are moderately sized (up to 4 layers of 2000 units); the behaviour of **MF** and **FF** under increasing depth, width, or dataset complexity is unknown, though prior work on local-loss methods [22] suggests scalability.
2. **Datasets.** MNIST, Fashion-MNIST, CIFAR-10, and CIFAR-100 are well-established but small-scale benchmarks. CIFAR-10 and CIFAR-100 inputs are flattened to 1-D vectors for MLP experiments, which eliminates spatial structure. Findings may not reflect behaviour on larger or more complex datasets.

3. **Hardware specificity.** All hardware results are specific to the NVIDIA A100-SXM4-40GB on the Athena cluster. Results may differ significantly on consumer GPUs (e.g., RTX 4090), TPUs, or edge hardware.
4. **Statistical robustness.** Three runs are insufficient to establish statistical significance at conventional thresholds, especially for small accuracy margins (e.g., +0.09 pp on MNIST); the MF MNIST advantage in particular should be interpreted with considerable caution.
5. **Hyperparameter search confound.** Each algorithm underwent independent Optuna tuning. Differences in performance may partly reflect better hyperparameter discovery rather than intrinsic algorithmic superiority. The hyperparameter search cost itself was not included in the reported training time and energy figures; for slow algorithms like FF, the 50-trial Optuna search overhead is substantial and should be considered in any total-cost assessment.
6. **Missing deep-supervision baseline.** MF introduces per-layer projection matrices and local cross-entropy losses, which both enable the BP-free training signal *and* add additional parameter capacity and supervision depth relative to the standard BP baseline. A BP network with equivalent per-layer auxiliary heads (BP-DS; “deep supervision” [14]) — adding projection heads of identical dimension with auxiliary cross-entropy losses at each layer and training end-to-end — would allow attribution of MF’s accuracy advantage specifically to the forward-only training regime. We were unable to complete this comparison within the present submission and include it as a priority item for future work. All efficiency comparisons (energy, time, memory) remain interpretable regardless.
7. **Inconsistent Fashion-MNIST MF result.** On Fashion-MNIST, MF incurred a 9.90% increase in energy relative to BP. This is inconsistent with the efficiency gains observed on CIFAR-10 and CIFAR-100. The cause is not fully understood; it may reflect dataset-specific convergence behaviour or hyperparameter interactions on the smaller network (2×1000 MLP). This result qualifies any claim of consistent efficiency superiority across all configurations.
8. **Scope of contribution.** This paper’s primary contribution is evaluative, not algorithmic; the benchmarking framework, fair comparison conditions, and hardware-level measurements are our contribution. The algorithms (FF, CaFo, MF) are previously published [6, 30, 4].

7 Conclusion

We presented a rigorous, hardware-validated benchmarking study of three forward-only, BP-free training algorithms, namely FF, CaFo, and MF, evaluated against fairly tuned BP baselines under a principled fairness framework on MLP and CNN architectures across four standard vision benchmarks.

On its native MLP architectures, MF consistently matched or surpassed BP in classification accuracy across all tested configurations, achieving up to 41% lower energy consumption and up to 34% faster training on CIFAR-10.

These findings challenge the assumption that global end-to-end optimisation is necessary for competitive accuracy on these architectures and establish MF as a practically viable training algorithm for this setting. By contrast, FF incurred prohibitive efficiency costs despite moderate accuracy gains, and CaFo faced an acute accuracy–efficiency trade-off that limited its practical utility without the expensive DFA pretraining stage.

Importantly, these results are specific to MLP architectures on standard, small-scale benchmarks on a single GPU platform. The finding that BP-free training can match or exceed BP’s accuracy and efficiency cannot be extrapolated to larger models, other architectures, or other hardware without further investigation. The absence of a deep-supervision BP baseline (BP-DS) is a notable gap that future work should address to isolate the algorithmic contribution of forward-only training.

Within its established scope, this work provides a reproducible benchmarking framework, hardware-level efficiency profiles, and an empirical re-evaluation of memory efficiency that together offer a more rigorous empirical foundation for evaluating BP-free methods than was previously available.

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7.1 Code Repository

To ensure full reproducibility, we provide the complete research artifacts associated with this work. This includes the full source code, all configuration files, selected hyperparameter YAML specifications, and per-run experimental logs. In addition, all appendix material previously included in the manuscript has been moved to the repository as an online appendix.

<https://github.com/Przemyslaw11/BeyondBackpropagation>

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